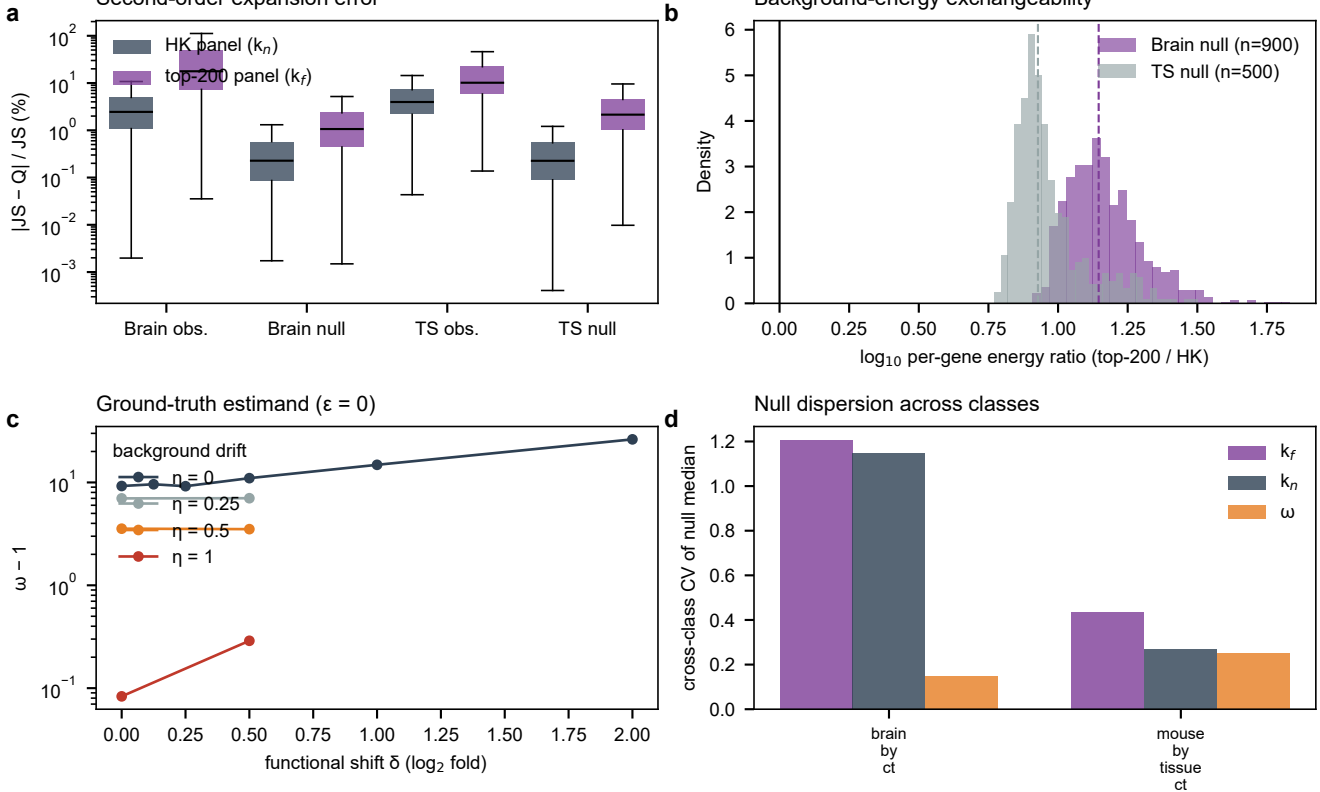

a
Second-order expansion error
HK panel (k_n)
top-200 panel (k_f)
|JS − QJ| / JS (%)
Brain obs.
Brain null
TS obs.
TS null
b
Background-energy exchangeability
Brain null (n=900)
TS null (n=500)
Density
log10 per-gene energy ratio (top-200 / HK)
c
Ground-truth estimand (ε = 0)
background drift
η = 0
η = 0.25
η = 0.5
η = 1
ω − 1
functional shift δ (log2 fold)
d
Null dispersion across classes
k_f
k_n
ω
cross-class CV of null median
brain by ct
mouse by tissue ct